A Multi-pronged Approach for Indoor Positioning with WiFi, Magnetic and Cellular Signals

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***Abstract*— As smartphones are increasingly popular, location-based services (LBSs) have become one of the crucial applications in daily lives. While outdoor localization is relatively easy leveraging GPS signals, localization in indoor environments is difficult due to the lack of GPS. Thus, due to the pervasive deployment of WiFi access points, there have been numerous studies on WiFi based indoor positioning.**

However, the multi-path fading of WiFi signals causes time-varying received signal strengths of WiFi signals, which leads to poor accuracy of WiFi localization. Moreover, WiFi scanning period, about 3~4 seconds in general smartphone, may provide poor quality of services in the context of refreshment interval.

Motivated by these limitations, we study the usability of two other sources that are currently available in smartphones: magnetic field strength and cellular signal strength, for indoor positioning purposes. Our preliminary measurements show that these two sources satisfy three properties needed for localization: time-invariance, location representativeness and universality. Because these three sensors have their own characteristics, some problems in single sensory data based localization could be complementarily overcome. We will show how to combine the three smartphone sensory data for indoor positioning based on each module's characteristics, and how much accuracy is achieved by the hybrid localization.

Keywords—Hybrid Positioning System, Indoor localization, LTE-based localization, Magnetic-based localization

I. Introduction

As recent smartphones are equipped with various sensors, many research groups are actively working on how to exploit the data from sensors and wireless communications to figure out user situations. In particular, finding the location of a user is extremely important since the location is often a key to estimate the user context, and to provide LBSs. In the case of outdoor positioning, GPS signals allow a smartphone to calculate its current location. In contrast, since GPS signals are not available in indoor environments, WiFi based fingerprinting is typically used because of its widespread deployment and easy measurements. The average localization error of WiFi based positioning technologies are usually reported about 2~3 meters in the literature, which is often carried out in well controlled and intensively tested environments. However, in reality, WiFi based positioning performance can be highly fluctuated by surrounding environments, such as people's density or mobility, the layout of obstacles, and even weather, all of which affect the Received Signal Strength (RSS) of WiFi signals. Thus, relying only on WiFi for localization is not a resilient approach.

There are other sensors and wireless communication interfaces in a smartphone, such as Bluetooth, NFC, Magnetic sensor, Accelerometer, Gyroscope, Cellular, etc. Bluetooth, and NFC could be used to indoor localization for the proximity detection, but they require additional cost to deploy Bluetooth devices and NFC readers/tags. Moreover, in the case of Bluetooth, it interferes with WiFi because they share the same frequency band.

Whereas, magnetic field information remains unaffected by surrounding environments, and it does not require the deployment of additional devices. We also find that the signal strengths from base stations (BSs) of cellular networks depend on the location and relatively stable compared to WiFi. That is, Reference Signal Received Power (RSRP) (reported by a smartphone) of a LTE BS is dependent on its position, and hence can represent the particular position.

The rest of this paper is organized as follows. In Section 2, we demonstrate that Magnetic and LTE signals satisfy three properties: time-invariance, spatial representativeness, and universality, which means that they are eligible to be used in indoor localization systems. In Section 3, we propose a hybrid indoor positioning technique using WiFi signals, Magnetic sensory data, and LTE signals, in order to overcome the fluctuation of the RSS of WiFi.

II. Feasibility of the use of Magnetic and Cellular Information

In order to use the magnetic field and cellular signal information for indoor positioning, we need to check whether the information satisfies the three properties: time-invariance, spatial representativeness, and universality. Time-invariance is important to localization system because the cost and efficacy of creating/maintaining a fingerprint map is highly related to temporal variability of signal. If the radio-map is not persistent as time goes by, the localization error may not be contained in a reliable manner. Spatial representativeness is also an essential property to identify the corresponding place to the sensory data. Some sensor information (e.g. temperature) may have little or no correlation with the position of a smartphone, which cannot be used for localization. Lastly, universality means that the sensor information should be available at any possible locations of users.

Magnetic field is already shown to satisfy the three properties [1]. But the eligibility test of LTE signals for indoor localization is not reported in the literature to the best of our knowledge. In LTE networks, there is a received signal indicator called RSRP. RSRP is the average received power over the resource elements

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which have cell-specific reference signals. Thus, we conclude that RSRP is eligible for indoor localization.

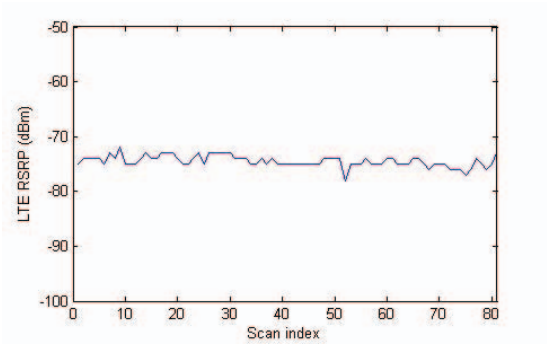

Figure 1: We measure LTE RSRP at a fixed position at 4 second intervals, which turn out to be relatively stable, compared with the measurement result in [7].

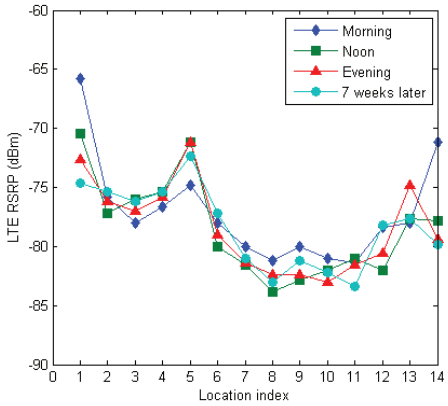

Figure 2: Average of LTE RSRP is plotted at different times at 14 different position, which demonstrates location dependency and time-invariance.

We measure the average RSRP of LTE signals at each position at different times. Figure 1 shows the LTE RSRP value at the same position measured 81 times (x-axis). The interval needed for refreshment (resetting the signal value) is about 3~4 seconds. From the 81 measurements, the average of the values is about -74.5 dBm, and the standard deviation is only about 1.00. From the observation, the RSRP values of LTE signals tend to be stable in a short timeframe. In Figure 2, we plot LTE RSRP values at 14 different positions (x-axis) in a long timeframe. We note that there is a distinct pattern at different positions but little diurnal difference in the RSRP values when we measure the values in the morning, noon and in the evening. Moreover, an almost similar pattern is observed after 7 weeks. Thus we decide to use the cellular signal information due to its time-invariance and representativeness. Furthermore, we can access the magnetic and cellular information everywhere in indoor environments in general. Therefore, we can conclude that the magnetic field strengths and LTE RSRP signals have a property of universality.

Notice that one of the lessons from our experiments is that the LTE signal measurement in smartphones sometimes has drawbacks. We experience often long delays (3~4 seconds) to retrieve the LTE RSRP values. Moreover, most of the current smartphones return a single value of RSRP (of the serving BS) even though there are multiple BSs around. This is in contrast to WiFi whose measurement is an N-dimension vector of RSS values of N access points (APs) around. Therefore, we seek to use not only the current value of LTE RSRP, but also its previous values for positioning.

III. Hybrid Positioning System

For localization purposes, each sensory data has strengths and weaknesses by their distinct characteristics. Thus, we could take advantage of their merits by fusing them based on their characteristics in contrast to the limitation of a single sensory-based localization system. At first, we present how each sensory module performs localization within our hybrid positioning system.

For the WiFi-based positioning scheme, we adopt the well-known grid-based fingerprinting approach like [3][5], due to its simplicity. We aggregate the RSS values collected at positions in each grid, and for each grid, compute the average RSS for each AP. In online phase, a user's position is estimated by computing a similarity score between the observed RSS values (of APs) of the user and those of each grid in the fingerprint map. Like [3], we exploit the RSS Euclidian distance for computing the similarity score, take k grids which have top-k similarity scores, and estimate a user's current position by k-NN (k-Nearest Neighbor) grid's average coordinate.

In case of the LTE based positioning module, we collect RSRP values like WiFi's offline phase. (Actually, we measure LTE and WiFi simultaneously.) Recall that LTE RSRP based localization has drawbacks like retrieving delay and one dimensional value. But fortunately, measured values of LTE RSRP reveal much less fluctuation compared to the WiFi case. As demonstrated in [6], positions of cellular base station are selected by optimum deployment strategy for avoiding the interference, so LTE signals also tend to be stable. This is useful if we can have the history of the measured RSRP values. Therefore, we consider user's previous RSRP value during a time window. We define an RSRP window with k previous measurements at i-th moment as follows.

$$RSRP\ \ windo\ \ w_k = \{RSRP_{i-k+1}, \cdots, RSR\,P_{i-1}, RSR\,P_i\}$$

And then, the values in the window are compared with those in the LTE fingerprint map. In the comparison process, we search the values in the map considering the user's moving distance. We can easily adjust the range of values in the map, which we call a candidate window, by inferring the moving distance based on the RSRP timestamps and floor plan's grids. The user's current position is estimated by finding the minimum RSRP "distance" between the observed RSRP window and candidate windows.

The fundamental problem of the use of magnetic sensory data in indoor space is its interference issue which is arisen by building structure or steel obstacles [1]. This interference is critical when we use smartphone as a compass, but it can also be exploited as an advantage when we leverages this interference as fingerprint patterns. One of the other advantages in magnetometer is that we can obtain a magnetic field strength in a short period (25 Hz in general smartphones). However, magnetic sensory data also has its limitation. While it consists of only 3 values (i.e., a 3D vector), the 3 axes are almost continuously changing

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as a user's phone moves, and we cannot figure out the directions of the 3 axes. Therefore, like prior approaches [1][4], we unite these magnetic values of the 3 axes values into a vector strength |m|, which turns out to be dependent on each location. And we collect the pattern/history of these magnetic strengths for a given timeframe continuously. Thus, we construct the magnetic fingerprint map. Then the user's observed magnetic strength pattern is compared with a range of candidate magnetic fingerprints. In the case of our experiments, the area of interest is actually one dimensional as shown in Figure 4. We will explain experimental settings in Section 4.

In online phase, we borrow the methodology in [1]. A user's observed magnetic strength pattern is defined as a magnetic window, and we compare the magnetic window with candidate magnetic fingerprints. As we shift the magnetic window along the magnetic fingerprints, we compute similarity scores to find the maximum likelihood position. For computing the similarity score, we exploit the Fast Dynamic Time Warping (FastDTW) method [2]. The DTW algorithm effectively finds similarity between two temporal sequences, each of which may be measured on different scale in terms of time or space. We use FastDTW since the user's speed may be different from the fingerprint collector's speed in offline phase.

So far, we have explained our localization approach from the perspective of each data source (WiFi, LTE, magnetic). From now on, we describe how we combine the above three localization modules for designing a hybrid positioning system. In case of WiFi, its signal strength fluctuates and is affected by its ambience. Thus, we should expect that the performance of positioning based on WiFi may vary with some substantial deviation. However, despite the precision concern, as many APs are typically located in indoor environments, multi-dimensional RSS values (of APs around) help us to "roughly" estimate the location of a user almost instantly. Thus, we use the WiFi localization module mostly for coarse-grained localization. That is, to reduce the search space in the area of interest (or the entire fingerprint map). From the experiments, we reach the conclusion that the search space can be reduced to within a radius of 10~13 meters from the estimated position of WiFi-only localization.

Then, we perform LTE positioning in the reduced search space. This could help improve the accuracy of the LTE-based positioning module because there could be many LTE RSRP candidates in the fingerprint map before user garners the sufficient history of LTE RSRP values. We could filter out false candidates by limiting the searching space.

In our experiments, leveraging the LTE positioning module, we can reduce the search space further down to an area whose radius is around 3~5 meters. Finally, as shown in Figure 3, we estimate the final user position by the magnetic-based positioning module within the reduced space. As we can sample the values of the magnetometer with a short scanning period (about 25Hz) in general smartphones, we can estimate the user's position not at the grid level, but at finer-grained level, almost continuously in the search space. Moreover, we repeatedly search the estimated position more frequently by the magnetic sensor, before the next search space is delivered from the other modules, thanks to the magnetometer's short scanning period.

Our system solves some problems of prior studies. Many research groups strive to improve the accuracy of WiFi based positioning systems, But in general smartphones, the WiFi scan period is about 3~4 seconds; thus, users are usually offered the relatively poor precision of localization services in the context of refreshing time. In contrast, thanks to the short magnetometer's scanning period, our system can update the estimate location at relatively short intervals.

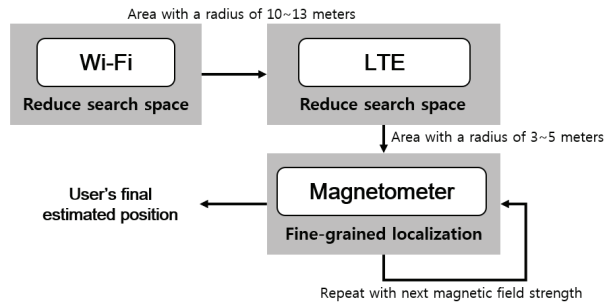

Figure 3: Overall structure of hybrid positioning system

Moreover, the LTE module along with the WiFi one which helps to reduce the search space can achieve more precise localization due to its signal stability. Lastly, the DTW algorithm, used in magnetic field, based similarity computation may require high computational cost. As our system provides reduced candidate area to calculate similarity of possible positions, this time complexity and power consumption issue may be relaxed.

IV. EVALUATION

All of the experiments are performed in building 301 at SNU campus, where the floor occupies 2,000m^2. We divide the area of interest into sub-areas, each being called a grid. The size of each grid is 3×3 sq. meters. In each grid, we collected 100 samples of WiFi signals and LTE RSRP signals. As to magnetic sensing, we collected magnetic field samples by taking a walk around the corridor as shown in Figure 4. We assume that a user walks at constant velocity and in forward direction along the corridor when we compare the user's sensory values with the fingerprint database of LTE and magnetic field signals. With exploiting the PDR scheme, these assumptions could be relaxed.

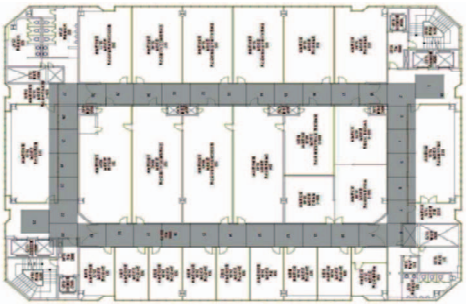

Figure 4: Floor plan of area of interest in Building 301 at SNU campus is shown. The measurements are performed only in the dark parts, whose length is around 120m long.

A. Accuracy of Individual Localization Modules

Figure 5 plots the CDF of the error distances estimated by each of WiFi, LTE and Magnetometer modules. WiFi's average and

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median of error distances are about 6.27 and 5.98 meters, respectively. Notice that its CDF increases steadily, which indicates almost uniform distribution of error distances due to signal fluctuation. This is not so desirable to be used for the localization purposes. Thus, we use WiFi to reduce the search space when we compare user's sensory data with the fingerprint DB. LTE's average and median of error distance are about 4.93 and 2.02 meters respectively. Its error distance is quite low compared to WiFi when the user's history data is accumulated enough. Each scanning/sensing happens at 4 second intervals. With that intervals, 4~5 scans (or 15~20 seconds in time) of the LTE RSRP history are known, the error is converged substantially. Thus, LTE is good to exploit for localization if its history is accumulated sufficiently.

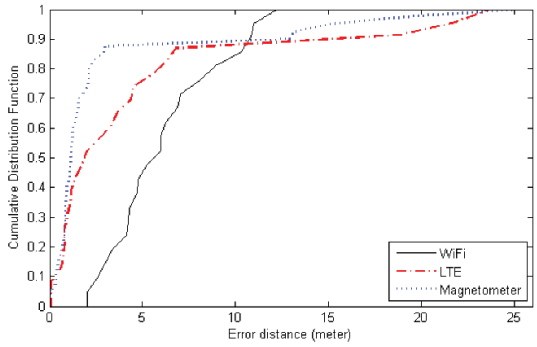

Figure 5: We plot the CDF of error distances of each localization module: WiFi, LTE, and Magnetometer.

Magnetic field based positioning is performed every 2 seconds in our system. It achieves the best performance compared with the other modules. Its average and median of error distances are about 2.54 and 1.13 meters, respectively. Note that the error distance here is possible only after accumulating enough magnetic samples as similar to the LTE case. Because it also requires the history of sample data for accurate positioning, the performance could be improved by reducing search space by WiFi and LTE.

B. Hybrid System Performance

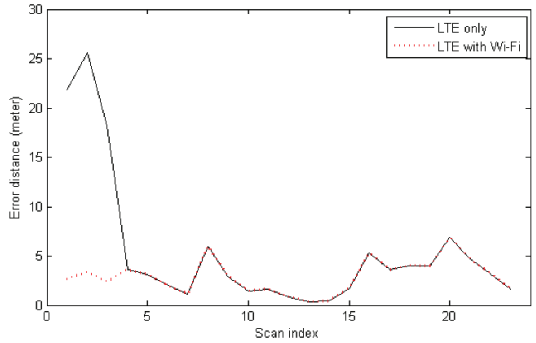

Figure 6: Error distances of the LTE localization module with and without WiFi are compared as a user walks along the corridor.

Figure 6 plots the positioning error distances of the LTE module with WiFi used to reduce the search space, compared with no WiFi case. In LTE only, its error distance is too high to use before convergence. Convergence time is around 20 seconds which means k = 4 samples. In case of LTE with WiFi, it shows lower error distance and convergence time is not needed. The average and median errors of LTE with WiFi are 2.91 and 2.89 meters, respectively.

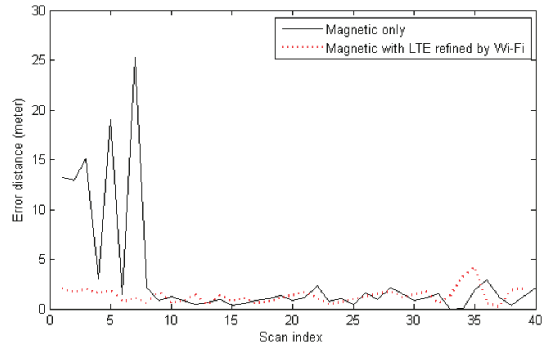

Figure 7: Magnetic positioning error distances with and without LTE and WiFi are plotted.

Figure 7 plots the magnetic module's error distances with and without LTE and WiFi. Before convergence time, magnetic only positioning error is so huge due to its 1-D fingerprint. The convergence time is around 20 seconds (2 seconds for each magnetic sensing in our experiments). However, with WiFi and LTE reducing the search space, the error is reduced even before convergence. The average and median of error distances are 1.30 and 1.22 meters, respectively.

V. Conclusion

This paper proposes and demonstrates a hybrid positioning scheme with WiFi, LTE and magnetometer. We investigate how to leverage each localization module to improve the final performance of the hybrid approach. WiFi RSS allows us to reduce the search space even for the cold start, and LTE helps us to reduce the search space even further due to stable LTE RSRP values. Finally, magnetic signals are exploited to improve the positioning performance not only in precision but also in position delay due to its fast scanning and more frequent availability.

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